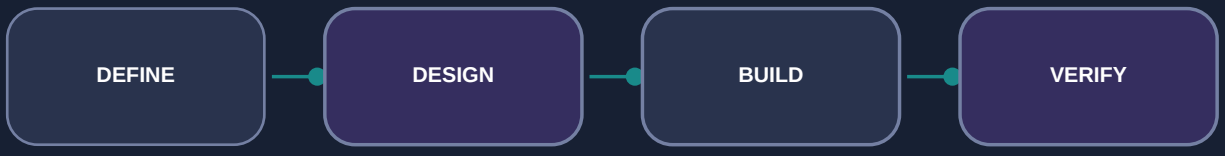
DOCUMENT PRD • Product Requirements Document

Product Requirements Document

Product vision, scope, functional requirements and acceptance criteria



Project name: Slotwise (provisional)

Proposed stack: React + Vite + Tailwind • Node.js + Express • Vercel

Public-source clean-room plan • illustrative graphics, not product screenshots

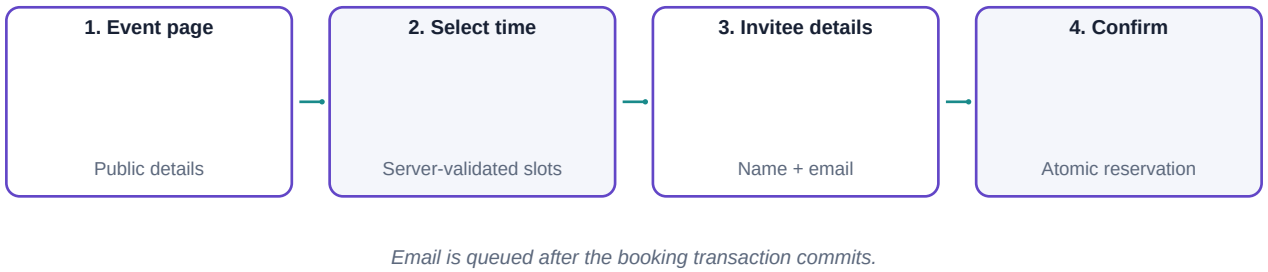
Visual summary

Product Requirements Document

PURPOSE

Define a focused scheduling MVP from event setup through reliable booking confirmation.

KEY VISUAL



Email is queued after the booking transaction commits.

REVIEW NOTE

Original content is retained after this summary; proposed visuals are illustrative, not real product screenshots.

DOCUMENT 01 OF 6

Product Requirements Document

A focused scheduling workflow for independent service providers

Project: Slotwise — provisional independent scheduling product

Reference: Calendly public website and help documentation; no private access.

Platform: Web | **Proposed frontend:** React + Vite + Tailwind CSS | **Backend:** Node.js + Express

Hosting preference: Vercel | **Database:** unresolved conflict between MongoDB preference and required PostgreSQL DDL

Inputs used / assumptions

App reference is Calendly (public website and help center). Target persona is an independent tutor, explicitly a provisional guess. The slice is event types, public booking, and booking confirmation. User complaints are not verified because no reviews or interviews were supplied.

Evidence discipline: Verified public capabilities are distinguished from design proposals and guesses. No review corpus, actual application screenshots, private account data, or production credentials were provided.

1. Executive summary

Slotwise is a proposed web application that lets a service provider define a session type, set weekly availability, share a public link, and receive confirmed appointments. The MVP optimizes for a single-host booking loop rather than full scheduling-platform parity.

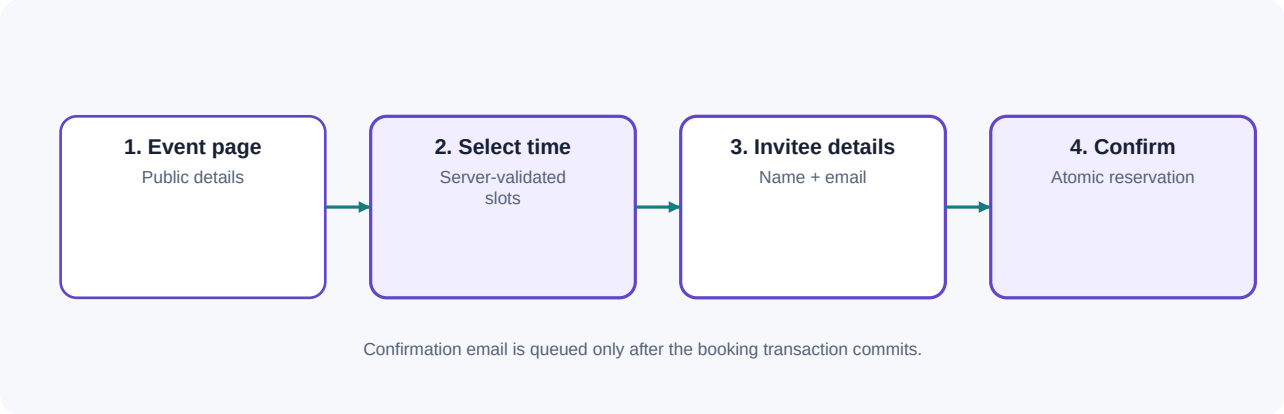


Figure 1. Core booking loop — illustrative process diagram, not an application screenshot.

2. Problem statement and product vision

Problem statement: Independent service providers need a reliable way to publish available appointment times and accept bookings without manually coordinating every time. The product hypothesis is that a narrow, clear flow can reduce confusion and scheduling back-and-forth. The amount of user pain is not yet measured.

Vision: Make a first booking easy to publish, easy to understand, and reliable under concurrent requests.

Core loop: Create event → define availability → publish link → invitee chooses a slot → submit details → confirm booking.

3. Scope and personas

Persona	Need	Pain hypothesis	Evidence / confidence
Independent tutor (primary)	Publish repeatable session types and let students book	Manual coordination takes time	Guess; medium confidence in use case, low confidence in differentiation
Student / invitee	Find and book an available time quickly	Unclear time zone or uncertain success state	Guess; low confidence until usability testing
Owner / operator	Prevent conflicting bookings and manage cancellations	Concurrent requests can create conflicts without strong integrity	Engineering risk; high confidence

4. Goals and non-goals

- Must: create and publish event types; define weekly availability; expose public booking page; validate slots server-side; confirm bookings; show owner booking list.
- Should: cancellation and rescheduling, clear time-zone display, understandable conflict recovery, accessible error states.
- Could: analytics, paid sessions, external calendar integrations.
- Non-goals: enterprise routing, marketplace, video platform, licensed content, original Calendly assets, private APIs, and a production zero-touch security response system.

5. Functional requirements

ID	Requirement	Priority	Acceptance criterion
FR-01	Owner can register, verify email, log in and log out	must	Unverified users cannot publish; session is revoked on logout
FR-02	Owner can create, edit and archive an event type	must	Validated event appears only to its owner until published
FR-03	Owner can define weekly availability and time zone	must	Invalid or overlapping local ranges are rejected

ID	Requirement	Priority	Acceptance criterion
FR-04	Visitor can view a published event and available slots	must	Unpublished events are not exposed
FR-05	Visitor can book without creating an account	must	Valid request creates exactly one booking
FR-06	System prevents overlapping confirmed bookings per host	must	Concurrent test leaves one valid booking at most
FR-07	System confirms booking and queues an email	must	Outbox record exists only after booking transaction commits
FR-08	Invitee can cancel or reschedule using a secure token	should	Invalid/expired tokens disclose no booking details
FR-09	Setup and conflict messages are clear and actionable	should	Usability test participants can recover from a conflict
FR-10	Paid bookings	could	Disabled until Stripe test-mode implementation is approved

6. Non-functional requirements

ID	Requirement	Target / validation
NFR-01	Performance	Core read API p95 < 500 ms in a representative test, excluding email delivery
NFR-02	Accessibility	WCAG 2.2 AA target; keyboard-only core loop and axe scans
NFR-03	Integrity	No double bookings in concurrency tests
NFR-04	Privacy	Minimize invitee data, define retention, support account deletion
NFR-05	Observability	Structured logs, request IDs, health endpoint, error monitoring
NFR-06	Responsive	Verified at 390px and 1440px

7. Positioning options and evidence

Option	Positioning	Evidence count	Decision
A	For independent tutors who want less scheduling back-and-forth, Slotwise provides a simple booking link and clear confirmation.	0 reviews / 0 review sources	Recommend; hypothesis to validate
B	For students who want to book a session without confusion, Slotwise makes available times and next steps clearer.	0 reviews / 0 review sources	Alternative; low confidence
C	For solo providers who want a focused scheduling tool, Slotwise handles event setup through confirmation.	0 reviews / 0 review sources	Alternative; medium use-case confidence

No customer complaint is asserted as fact. Ranked pain points—setup friction, time-zone uncertainty and unclear conflict recovery—are research hypotheses until supported by reviews or interviews.

8. Feature prioritization and proposed fixes

Feature	Area	Priority	Original capability?	Slotwise implementation / fix
Event type setup	Setup	must	Yes, documented broadly	Fresh form, guided defaults
Weekly availability	Setup	must	Yes, documented broadly	Explicit zone and validation
Public booking page	Booking	must	Yes, documented broadly	Original page and copy
Slot selection + atomic reservation	Booking	must	Yes, core scheduling job	Database-level overlap protection
Confirmation	Booking	must	Yes, documented	Original success state with exact time zone
Owner booking list	Management	must	Yes, broadly	Filter by date and status
Clear conflict recovery	UX fix	should	Not established	Refresh availability and explain next step
Policy preview	UX fix	should	Not established	Show cancellation terms before booking
Tutor-focused presets	UX fix	could	Not established	Original duration/session presets

9. Business rules and constraints

- Only published, non-archived events are bookable publicly.
- Invitees do not need accounts for the MVP.
- Booking start must precede booking end; duration must match the event configuration.
- All stored appointment instants use UTC; event and invitee display zones use IANA time-zone names.
- A confirmed booking may not overlap another confirmed booking for the same host.
- Emails are queued after the booking transaction commits; retries must be idempotent.
- Payment collection and external calendar synchronization are not MVP dependencies.

10. Out of scope / cannot be cloned

Item	Decision	Reason
Calendly code, HTML/CSS/JS, logos, icons, fonts and copy	skip	Clean-room requirement and proprietary assets
Existing Calendly users/network	skip	Cannot reproduce another product's user relationships
Partner deals and licensed content	skip	Requires independent agreements
Regulated licenses and hardware	skip	Not provided and not assumed
Team routing and enterprise features	skip for MVP	Outside the selected core slice
External calendar conflict checking	defer	Requires approved official API/OAuth integration

11. Pricing hypothesis and KPIs

Tier	Proposed price	Proposed scope
Free	■0	One active event type and core booking
Solo	■299/month	Multiple event types and reminders
Plus	■599/month	Advanced customization and basic analytics

Pricing is a hypothesis, not a competitor price claim. Consider an annual offer equivalent to two months free only after willingness-to-pay testing. If paid billing is introduced, provide clear renewal notices and self-service cancellation.

KPI	Initial target	Measurement
Event setup completion	≥ 80%	Completed setup / setup started
Booking completion	≥ 70%	Confirmed booking / booking flow started
Double bookings	0	Concurrency tests and production incidents
Confirmation queue success	≥ 99%	Accepted jobs / eligible bookings
API read p95	< 500 ms	Performance test excluding email
Open S1/S2 at release	0	Defect tracker

12. Risks, size and evidence

Risk	Impact	Mitigation
Double booking	Critical	Atomic transaction, unique/exclusion constraint, concurrency tests
DST/time-zone errors	High	IANA zones, UTC storage, DST boundary tests
Weak differentiation	High	Interview target users and measure task success
Dataset of complaints unavailable	Medium	Do not claim user pain; collect public reviews/interviews
Database decision unresolved	High	Approve PostgreSQL or redesign the required integrity strategy for MongoDB

Size estimate: L. Hard parts are booking integrity, time-zone correctness, secure management links, email retries, and deciding between the MongoDB preference and the required PostgreSQL schema. References: Calendly official website and help documentation listed in the source register in Document 6.